

Sai Karthik Gutha

Email: saikgutha9@gmail.com

Mobile: +18482565355

Data Scientist

PROFESSIONAL SUMMARY:

- Data Scientist with 3+ years of experience in banking and healthcare domains; developed predictive models, anomaly detection systems, and real-time analytics using Python, SQL, PySpark, R, and Scikit-learn.
- Built fraud detection and credit risk models using decision trees, XGBoost, and time-series data, improving loan default accuracy by 18% and reducing transaction fraud false positives by 15%.
- Developed GPT and LangChain-powered LLM assistants for compliance Q&A workflows and document summarization, reducing analyst effort by 30% and accelerating regulated industry review times by 40%.
- Engineered NLP pipelines for EHR data using NER and regex methods, improving ICD code tagging by 25% and enabling faster clinical documentation and billing accuracy for healthcare clients.
- Built hospital readmission risk prediction systems and survival models, reducing readmissions by 20%, enabling timely patient interventions, and improving resource allocation through automated risk stratification.
- Automated ML pipelines with Databricks, Airflow, and SageMaker; cut training cycle times by 35%, supported hourly model retraining, and optimized Spark performance to reduce pipeline runtimes by 60%.
- Designed anomaly detection and trust scoring systems for real-time pipelines; flagged data integrity issues early, improving downstream data quality and preventing late-stage model errors.
- Created interactive dashboards with Tableau and Power BI to visualize KPIs, anomaly scores, and model insights, improving stakeholder decision-making efficiency by 40% across technical and non-technical teams.
- Built stock forecasting models using sentiment data, technical indicators, and ML algorithms with 80% signal precision; applied A/B testing and hypothesis-driven modeling to boost reliability and confidence.
- Practiced Agile development with Git, Docker, and CI/CD workflows; mentored junior analysts, improved team model delivery by 25%, and ensured reproducibility across distributed, cloud-based data environments.

TECHNICAL SKILLS:

- **Programming Languages** - Python, R, SQL, PySpark, Bash
- **Machine Learning & AI** - Scikit-learn, XGBoost, Random Forest, Logistic Regression, PCA, SHAP, LIME, A/B Testing, Hypothesis Testing, Model Drift Detection
- **Deep Learning & Generative AI** - TensorFlow, Transformers, Hugging Face, LangChain, GPT (LLMs), Prompt Engineering, Document Q&A, RAG Pipelines
- **Natural Language Processing (NLP)** - NER, Regex, Text Classification, Clustering, Tokenization, Text Summarization
- **Data Engineering & ETL** - Azure Data Factory, PySpark, SQL (PostgreSQL, SQL Server), MongoDB, AWS Glue, Apache Spark, Delta Lake, Airflow
- **Cloud Platforms** - Microsoft Azure (Data Factory, Azure ML), AWS (SageMaker, S3, Glue)
- **Big Data Ecosystem** - Databricks, Apache Spark (SQL & Streaming), Hive, Kafka
- **Data Visualization & Reporting** - Power BI, Tableau, Matplotlib, Seaborn, Plotly
- **DevOps & MLOps** - Git, GitHub, Docker, MLflow, CI/CD Pipelines, Model Deployment, Reproducibility
- **Collaboration & Workflow Tools** - Confluence, Slack, Notion, Jira, Agile
- **Data Formats & Modeling** - Parquet, JSON, CSV, Star Schema, Snowflake Schema, Data Normalization, Imputation

PROFESSIONAL EXPERIENCE:

JP Morgan Chase

Aug 2024 – Present

Data Scientist

Responsibilities:

- Built fraud detection and anomaly models using time-series analysis, PySpark, and GPT embeddings, reducing false positives by 15% and uncovering irregular entries across financial ledgers in real time.

- Designed credit risk prediction pipelines with decision trees and logistic regression, boosting delinquency detection accuracy by 18% and enabling proactive, risk-based pricing strategies for financial product underwriting.
- Developed LLM-powered chatbot using GPT and LangChain to automate compliance document analysis, reducing manual review time by 30% and increasing policy adherence across operational departments.
- Created churn prediction and customer segmentation models using ensemble methods, contributing to a 12% customer retention boost via targeted campaigns across retail banking and credit card portfolios.
- Automated model training using Airflow, Databricks, and AWS SageMaker; reduced refresh time from weekly to daily, scaling ingestion of 10M+ records via PySpark, Glue, and S3.
- Conducted advanced feature engineering on credit scores, utilization, and transaction patterns; improved F1-score by 14% and built interpretable models using SHAP and LIME for audit transparency.
- Built Power BI dashboards that visualized model outputs into actionable insights for fraud, risk, and loan departments, enabling data-driven executive decision-making and faster intervention cycles.
- Led model drift monitoring with Python, MLFlow, and AWS Glue; maintained performance across quarters and collaborated with MLOps teams to reduce deployment time by 45% using CI/CD.
- Integrated NLP-driven classification models for customer support and compliance; improved intent routing accuracy by 22% and reduced audit breaches by 25% through LLM-enhanced risk scoring workflows.
- Mentored analysts on TensorFlow, scikit-learn, and LangChain; scaled departmental modeling capacity by 3X and uncovered novel fraud clusters through GPT-powered transaction description clustering across AWS S3 datasets.

Value Labs

Aug 2022 – Apr 2023

Jr. Data Scientist

Responsibilities:

- Designed predictive fraud models using XGBoost and logistic regression on healthcare claims, reducing fraudulent payouts by 18% and increasing reimbursement accuracy across payers using structured medical data in Azure environments.
- Built scalable NLP pipelines using NER and regex for ICD code extraction from EHRs, improving billing precision by 25% and enhancing automation in high-volume clinical documentation workflows.
- Conducted feature analysis to isolate diagnostic frequency signals; applied SHAP for model explainability, boosting clinical stakeholder trust and enabling compliance audit readiness through interpretable visuals and documentation.
- Created anomaly visualizations in Power BI for fraud pattern detection; enabled real-time monitoring and investigation by operations teams, increasing responsiveness to suspicious billing behavior across hospital groups.
- Engineered preprocessing templates using Pandas and Scikit-learn, cutting wrangling time by 40%; used Azure Data Factory and PySpark to reduce ETL pipeline latency by 50% from raw CSV sources.
- Tuned ML models via GridSearchCV and random search; validated with ROC-AUC and confusion matrix metrics to reach over 85% model accuracy and ensure reliable production predictions.
- Applied PCA for dimensionality reduction on EHR vectors, improving model runtime by 30% while retaining clinical signal strength; monitored drift using batch-wise comparison plots for ongoing reliability.
- Supported deployment of readmission risk prediction model, reducing hospital readmissions by 20%; documented lifecycle in Git and Confluence to boost onboarding efficiency by 3X across data science teams.

IHub Data

Oct 2021 – Jul 2022

Data Science Intern

Responsibilities:

- Developed cancer risk classifiers using Random Forest and logistic regression on structured clinical data, achieving 85% test accuracy and boosting recall by 12% through engineered interaction terms in Python.
- Cleaned and imputed clinical records from MongoDB and SQL sources, reducing missing data by 22%; applied normalization techniques and automated weekly syncs to ensure freshness in prediction experiments.

- Conducted exploratory analysis on risk factors like alcohol and tobacco; designed histograms, box plots, and heatmaps to visualize trends, supporting data validation for model-ready cancer datasets.
- Built survival models to estimate post-treatment longevity; supported oncologists with improved patient stratification and generated ROC-based performance reports to drive clinical research grant decisions.
- Designed Tableau dashboards for visualizing treatment impact and cancer risk heatmaps; delivered accessible visuals that enhanced collaboration between data scientists and non-technical research teams.
- Created reusable notebook templates and user-friendly data query tools, streamlining analysis and ensuring consistent machine learning workflows across researchers and interns in the medical analytics division.
- Built cross-validation pipelines using StratifiedKFold to maintain label balance; collaborated with engineers to align schemas with modeling goals, reducing iteration cycles by 25% in training loops.
- Performed literature reviews to align modeling with evidence-based protocols; helped standardize practices across teams and validated techniques using precision, recall, and performance metrics for every pipeline iteration.

Educational Details:

Master of Science in Computer Science - Wilmington University, Delaware